Additional Experiments

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We present additional experiment results which incorporate new benchmark algorithms.

1 Experiment Setup

Dataset and Testing Problems. We test on the same datasets as in the main paper.

Algorithm Benchmark. In the additional experiments, we incorporate the following new benchmarks

- GD-LS. Gradient descent with Armijo line-search.
- AGD-CVX-LS. Accelerated gradient descent with Armijo line-search applied to the descent step.
- ADPGD. Adaptive gradient descent without descent [1].
- ADPGDACC. ADPGD with acceleration [1].

Algorithm Configuration.

- Line-search condition is taken to be standard Armijo rule:

$$f(x-\alpha\nabla f(x)) \le f(x) - \alpha c\|\nabla f(x)\|^2,$$

  where $c = 10^{-4}$ and α is obtained using back-tracking. Whenever the test passes without back-tracking, the initial back-tracking stepsize α_0 is updated by $\alpha_0 \leftarrow 1.2\alpha_0$.
- We search the λ_0 parameter in ADPGD [1] (**Algorithm 1**) in the range $\{0.1/L, 1/L, 10/L, 100/L\}$.
- ADPGDACC uses the default heuristic parameters provided by [1].

Since it is not standard to equip GD-HB with line-search, we do not include it in the experiments.

Other Configurations. All the rest of the configurations remain the same as the main paper.

2 Experiment Results

Table 1 summarizes the updated statistics for thew new algorithm benchmarks

Algorithm/Problem	SVM (33)	Logistic Regression (33)
GD	5	2
GD-LS	9	9
GD-HB	9	7
AGD-CVX	8	3
AGD-CVX-LS	10	9
Adam	26	11
AdaGrad	9	8
ADPGD	9	9
ADPGDACC	11	11
L-BFGS-M1	13	11
L-BFGS-M3	20	14
L-BFGS-M5	26	16
L-BFGS-M10	31	18
BFGS	32	26
HDM-Best	32	21

Table 1. Number of solved problems for each algorithm

We observe that although line-search improves the convergence behavior of the baseline algorithms, the algorithms with a single stepsize seem less competitive with algorithms with a diagonal preconditioner when a *moderate-to-high-accuracy* solution is needed. Here our requirement $\|\nabla f(x)\|_\infty \leq 10^{-4}$ (which corresponds to function value gap of $10^{-7} \sim 10^{-9}$) is often too high for typical first-order methods.

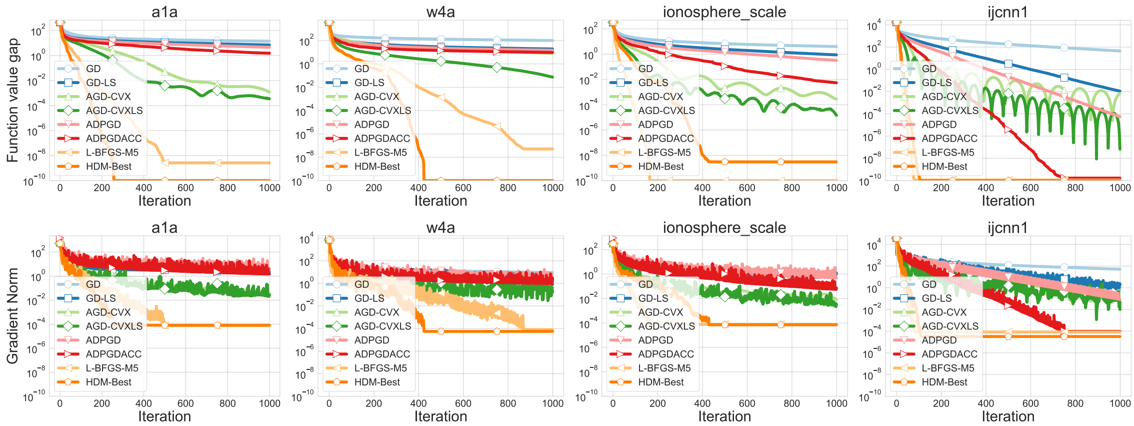

Figure 1. Performance of the algorithms on SVM problems

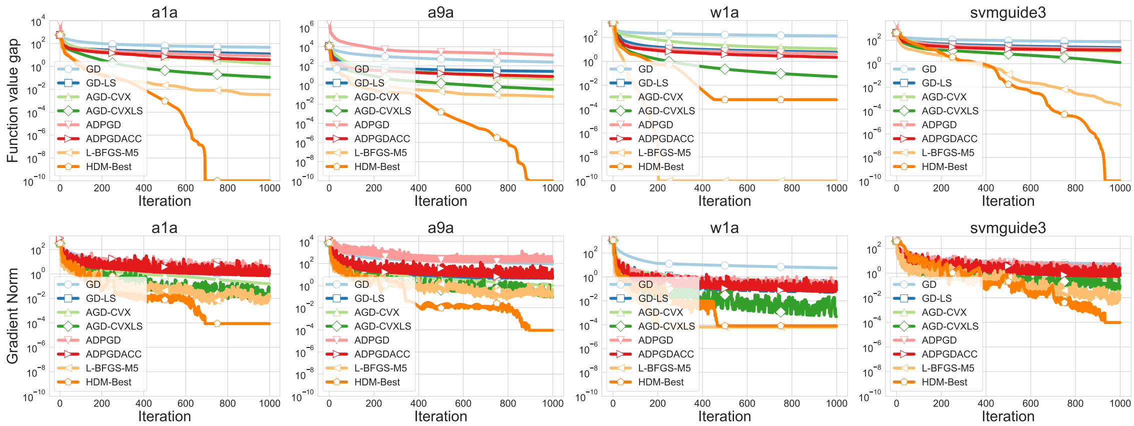

Figure 2. Performance of the algorithms on logistic regression problems

References

[1] Yura Malitsky and Konstantin Mishchenko. Adaptive gradient descent without descent. In *Proceedings of the 37th International Conference on Machine Learning*, ICML'20. JMLR.org, 2020.